

Chinese Elm

Ulmus parvifolia

Species background: Native to China, Korea, Japan, and Vietnam; introduced to the US in the early 1900s and increasingly planted as a naturally disease-resistant substitute after American elm losses. Its bark peels away in irregular patches of gray, green, orange, and brown, creating a mottled, camouflage-like pattern -- a completely different look from the deeply furrowed bark of American elm.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Chinese Elm: Identify Chinese elm's puzzle-piece bark pattern.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons (brown, tan, gray, orange).

Teacher background

My bark looks like a puzzle of gray, tan, and orange patches! I'm related to the American elm, but I don't get sick as easily.

Procedure

- 1 Engage: Ask if students have ever seen a puzzle piece pattern in nature.
- 2 Explore: Look at the bark's patchwork colors together.
- 3 Explain: Explain that this bark pattern is one of the easiest ways to spot this tree.
- 4 Art: Find the smallest leaf you can and trace it.
- 5 Evaluate: Student can describe the bark as a colorful patchwork or puzzle pattern.

Discussion questions

What other patterns in nature look like a puzzle or patchwork?

Assessment

Correctly describes the bark's patchwork appearance.

Extension

Find another patchwork-barked tree, like London planetree, and compare the patterns.

Chinese Elm: Compare Chinese elm's bark and leaf size to American elm's.

Standards connection: NGSS 3-LS4-4; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

Look for a patchwork of gray, tan, and orange bark colors. This tree's small leaves and different bark help tell it apart from American elm.

Procedure

- 1 Engage: Ask why two trees in the same family might look so different from each other.
- 2 Explore: Compare this tree's leaf size to a picture of an American elm leaf.
- 3 Explain: Discuss how this tree became a popular replacement after American elm losses.
- 4 Art: Trace the puzzle-piece pattern of the bark and color each 'puzzle piece' a different shade of brown, tan, and gray.
- 5 Evaluate: Student names one difference between this tree and American elm.

Discussion questions

Why might a city plant a related but different species instead of trying to bring back the exact original tree?

Assessment

Correctly names at least one distinguishing difference from American elm.

Extension

Look for an American elm nearby to compare directly, if one exists.

Chinese Elm: Explain disease-resistant substitution as an urban forestry strategy.

Standards connection: NGSS MS-LS2-1; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, cut-paper mosaic materials.

Teacher background

Chinese elm's exfoliating bark and natural resistance to Dutch elm disease made it a common replacement species after American elm losses -- connects directly to the elm case study covered elsewhere in this curriculum.

Procedure

- 1 Engage: What would you look for in a replacement species after losing most of one kind of street tree?
- 2 Explore: Research why Chinese elm was chosen as a common American elm substitute.
- 3 Explain: Discuss disease resistance as a key criterion in substitute species selection.
- 4 Art: Create a mosaic-style artwork using cut paper in bark-inspired colors -- gray, tan, orange, cream -- to recreate the patchwork bark pattern.
- 5 Evaluate: Write a paragraph explaining why disease resistance made this tree a popular substitute species.

Discussion questions

What other qualities, besides disease resistance, should a city consider before adopting a substitute tree species widely?

Assessment

Paragraph correctly explains disease resistance as the substitution rationale.

Extension

Compare Chinese elm's substitution story to zelkova's, which serves a similar role.

Chinese Elm: Evaluate tradeoffs in urban forestry risk management when selecting disease-resistant substitute species.

Standards connection: NGSS HS-LS2-7; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, camera or drawing materials for a value study.

Teacher background

Chinese elm's natural resistance to Dutch elm disease made it a common post-epidemic replacement species; research how urban foresters balance disease resistance against invasiveness concerns when selecting a substitute species, and evaluate whether this species carries any documented ecological tradeoffs of its own.

Procedure

- 1 Engage: Is a 'safe' substitute species ever completely risk-free, or does every choice carry some tradeoff?
- 2 Explore: Research any documented ecological concerns associated with Chinese elm plantings.
- 3 Explain: Evaluate the tradeoffs urban foresters weigh between disease resistance and other ecological factors.
- 4 Art: Photograph a section of bark and create a value study -- a grayscale drawing showing only light and shadow -- capturing how the patchwork pattern reads without color.
- 5 Evaluate: Submit the tradeoff evaluation and the grayscale value study.

Discussion questions

How should urban foresters weigh a known, documented risk (disease susceptibility) against an uncertain, less-studied one (potential future ecological impact)?

Assessment

Rubric covering research depth on tradeoffs and technical quality of the value study.

Extension

Compare Chinese elm's substitution tradeoffs to callery pear's -- a species introduced for disease resistance that later became invasive.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.